ALVA'S
Education Foundation®

Thought-Controlled Bionic Hand With Vision and Sensory Feedback

Akshith JK – 4AL23IC001
Leevan Michael Vaz – 4AL23IC022
Mohammed Nishmal – 4AL23IC025
Pavan Kumar – 4AL23IC032

Guide: Prof. Jyotibha R Chinchankar
Assistant Professor
Dept. of CSE (IoT, Cybersecurity including Blockchain)

Alva's Institute of Engineering & Technology, Moodbidri | April 2026

Contents

1 Introduction

2 Project Motivation

3 Literature Review

4 Problem Statement & Objectives

5 Proposed Methodology

6 Hardware & Software Requirements

7 Expected Outcomes

8 References

Introduction

- **Domain:** Prosthetics & Brain-Computer Interfaces (BCIs) — advancing to restore lost limb function and improve amputees' quality of life.
- **Technology:** Integration of EEG, sEMG, computer vision (YOLOv8), IoT sensors, and 3D-printed mechanics into a single intelligent system.
- **Challenges:** Existing prosthetics cost ₹5–50 lakhs, lack sensory feedback (pressure/temperature), and have no vision intelligence — leading to up to 75% abandonment rates.
- **Our Approach:** A multi-modal, affordable (<₹30,000) prosthetic hand combining neural control, stereo vision, and haptic feedback.

Project Motivation

75%

Abandonment Rate

of prosthetic users discontinue due to discomfort and lack of sensory feedback

2.5M+

Amputees in India

require affordable and functional prosthetic solutions urgently

₹5–50L

Avg. Cost of Prosthetics

making advanced devices inaccessible to middle-class amputees in India

0

Affordable Smart Prosthetics

combining EEG + sEMG + vision + haptic feedback exist at low cost

Literature Review

Table: Summary of Literature Review

Paper / Author	Method	Advantages	Limitations
Islam et al. (2025) – Acta Mechanica	sEMG + FSR + MLX90614 + 3D hand	98.2% motion accuracy	No EEG, no vision
NIH Study (2024) – EEG hand movement	EEG + SVM + Bayesian opt.	94.39% accuracy	Requires headset
Univ. of Utah (2025) – Nature Comms	Shared autonomy + vision	Autonomous grasp planning	Expensive hardware
Xie et al. (2024) – Frontiers in Neuro.	Peltier thermal feedback	Hot/cold sensation (15–40°C)	No pressure feedback
Johns Hopkins (2024) – Soft-rigid hand	Hybrid soft-rigid + silicone	Realistic human-like feel	No brain control
IIT/IEEE (2025) – Stereo + YOLO	Stereo vision + YOLOv8	Real-time object recognition	No prosthetic integration
S.A.P.I.E.N.S. (2026) – EEG+EMG	EEG + EMG hybrid control	Proves dual control feasibility	Not a complete product

Problem Statement & Objectives

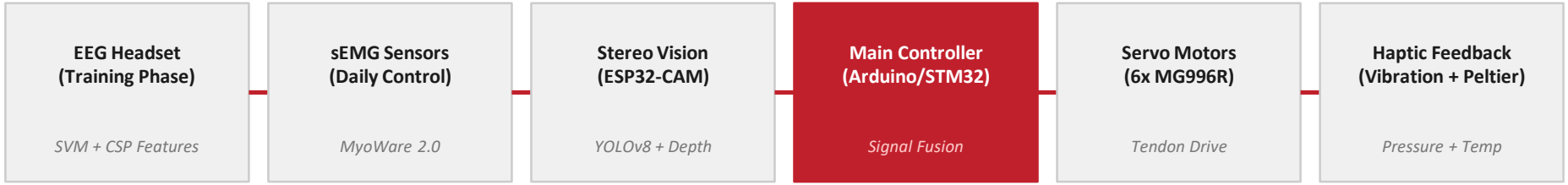
Problem Statement

Existing prosthetics cost ₹5–50 lakhs, lack sensory feedback (pressure/temperature), have 75% abandonment rates, no vision-based intelligence, and no bilateral coordination — all at an affordable price point. No low-cost system integrates EEG training, sEMG control, stereo vision, and haptic feedback into a unified solution.

Objectives

1. Design a low-cost bionic hand (<₹30,000) using 3D printing with five functional fingers and flexible TPU joints.
2. Implement EEG-based neural training (~94% accuracy) and real-time sEMG control for daily use — no headgear needed.
3. Integrate stereo vision (dual ESP32-CAM + YOLOv8) for object recognition and autonomous grasp planning.
4. Provide multi-modal sensory feedback: grip pressure (FSR), temperature (MLX90614), haptic vibration + Peltier thermal.
5. Enable bilateral coordination with natural arm via MPU6050 IMU and ESP-NOW wireless (<10ms latency).

Proposed Methodology



↕ **Bilateral Coordination: MPU6050 IMU on real arm → ESP-NOW (<10ms) → Prosthetic mirrors natural arm movement**

Key Technical Methods

- Signal Processing: Bandpass (7–35Hz), notch (50Hz), ICA artifact removal
- Feature Extraction: CSP + mu (8–13Hz) & beta (14–30Hz) band power
- Machine Learning: SVM with Bayesian optimization → ~94% accuracy
- Computer Vision: YOLOv8 object detection + SIFT stereo depth mapping
- Mechanical: 3D printed PLA/PETG skeleton, TPU flex joints, Ecoflex silicone skin

Hardware & Software Requirements

Hardware Requirements

Component	Qty	Cost (₹)
ADS1299 EEG + BioAmp Band	1	3,000
MyoWare 2.0 EMG Sensors	3	13,797
ESP32-CAM (OV2640)	2	3,000
FSR402 Pressure Sensors	5	825
MLX90614 Temp Sensor	1	500
MPU6050 IMU	2	500
MG996R Servo Motors	6	1,320
Arduino Uno R3 + ESP32	–	1,400
PLA + TPU + Ecoflex Silicone	–	4,200
Li-Po Battery + Regulators	–	1,700
TOTAL		**~₹30,500**

Software Requirements

Arduino IDE	Microcontroller programming
Python + MNE	EEG signal processing
scikit-learn	SVM classifier training
OpenCV + YOLOv8	Computer vision & object detection
Edge Impulse	Embedded ML deployment
ESP-NOW Library	Wireless communication
Fusion 360 / Solid Edge	CAD design for 3D printing

Expected Outcomes

01 Functional Prototype

5-finger bionic hand with tendon drive, TPU flex joints & silicone skin — total cost below ₹30,000.

02 Neural + EMG Control

EEG trains once; daily use with shoulder EMG sensors only — no headgear required.

03 Stereo Vision

Recognizes objects (cup, egg, laptop) and auto-selects optimal grip type via YOLOv8 + depth mapping.

04 Multi-modal Feedback

Feel grip pressure via vibration motors; detect hot/cold via Peltier element (15–40°C range).

05 Bilateral Coordination

Prosthetic mirrors natural arm — arm swing, lifting, clapping — via MPU6050 + ESP-NOW.

06 Societal Impact

Affordable solution for middle-class amputees in India; deployable as a kit for rehabilitation centers.

References

[1] Islam, S. et al. (2025). Electromyography Signal Operated Bionic-Hand. Acta Mechanica et Automatica, 19(4).

[2] NIH Study (2024). Brain synergy features for EEG-based hand movement classification. NIH Database.

[3] University of Utah / Nature Communications (2025). Shared autonomous control for prosthetic hands.

[4] Xie, A. et al. (2024). Hybrid sensory feedback system for thermal nociceptive warning in prosthetic hand. Frontiers in Neuroscience.

[5] International Journal of Bioprinting (2023). Metamaterial flexure joints for bionic hands.

[6] Johns Hopkins University (2024). Hybrid soft-rigid prosthetic hand design. Conference.

[7] IEEE/RSJ IROS (2024). Multiple Proactive Behavior Learning for mobile robots. Conference.

[8] IEEE (2023). Categorization of control strategies for bilateral tasks in arm prosthetics. Conference.

[9] Lauretti, C. et al. (2016). EMG + inertial sensor fusion for multi-DOF control. Conference.

[10] S.A.P.I.E.N.S. Project (2026). EEG+EMG prosthetic, Politecnico di Milano. Project Report.

ALVA'S
Education Foundation®

Thank You!

Thought-Controlled Bionic Hand With Vision and Sensory Feedback

Akshith JK | Leevan Michael Vaz | Mohammed Nishmal | Pavan Kumar

Guide: Prof. Jyotibha R Chinchankar | Dept. of CSE (IoT, Cybersecurity including Blockchain)

Alva's Institute of Engineering & Technology, Moodbidri